

HDOC - Day 19 Activity

C Programming Assignment

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

1. What are SGPA and CGPA?

SGPA (Semester Grade Point Average) tells you how well you performed in one semester. **CGPA** (Cumulative Grade Point Average) tells you how well you have performed overall, across all semesters so far. Every course has a **credit** (its weight) and a **grade** (O, A, B, C, D, E or P). Each grade has a fixed grade point (GP):

Grade	O	A	B	C	D	E	P
Grade Point	10	9	8	7	6	5	4

C Program

```
#include <stdio.h>
int getGP(char grade) {
    switch (grade) {
        case 'O': return 10;
        case 'A': return 9;
        case 'B': return 8;
        case 'C': return 7;
        case 'D': return 6;
        case 'E': return 5;
        case 'P': return 4;
        default: return -1; /* invalid */
    }
}

int main() {
    int global_id, m, n, semester, course, code, credit, GP;
    int semester_TC, semester_TGC, cumulative_TC, cumulative_TGC;
    char grade;
```

```

float SGPA, CGPA;
do {
printf("Enter Global ID: ");
scanf("%d", &global_id);
if (global_id <= 0)
printf("Error: Global ID must be positive. Please re-enter.\n");
} while (global_id <= 0);
do {
printf("Enter number of semesters (m): ");
scanf("%d", &m);
if (m <= 0)
printf("Error: Number of semesters must be positive. Please re-enter.\n");
} while (m <= 0);
cumulative_TC = 0;
cumulative_TGC = 0;
for (semester = 1; semester <= m; semester++) {
printf("\n----- Semester %d ----- \n", semester);
semester_TC = 0;
semester_TGC = 0;
do {
printf("Enter number of courses (n) for semester %d: ", semester);
scanf("%d", &n);
if (n <= 0)
printf("Error: Number of courses must be positive. Please re-enter.\n");
}
while (n <= 0);
for (course = 1; course <= n; course++) {
printf("\n Course %d of Semester %d:\n", course, semester);
do {
printf(" Enter course code: ");
scanf("%d", &code);
if (code <= 0)
printf(" Error: Course code must be positive. Please re-enter.\n");
}
while (code <= 0);

```

```

do
    {
    printf(" Enter credit: ");
    scanf("%d", &credit);
    if (credit <= 0)
    printf(" Error: Credit must be positive. Please re-enter.\n");
    while (credit <= 0);
    do {
    printf(" Enter grade (O,A,B,C,D,E,P): ");
    scanf(" %c", &grade);
    if (grade >= 'a' && grade <= 'z')
    grade = grade - 'a' + 'A';
    GP = getGP(grade);
    if (GP == -1)
    printf(" Error: Invalid grade. Please re-enter.\n");
    }
    while (GP == -1);
    semester_TGC = semester_TGC + (GP * credit);
    semester_TC = semester_TC + credit;
    }
    SGPA = (float) semester_TGC / semester_TC;
    printf("\n Semester %d SGPA = %.2f\n", semester, SGPA);
    cumulative_TGC = cumulative_TGC + semester_TGC;
    cumulative_TC = cumulative_TC + semester_TC;
    }
    CGPA = (float) cumulative_TGC / cumulative_TC;
    printf("\n===== RESULT =====\n");
    printf("Global ID : %d\n", global_id); printf("Semesters (m) : %d\n", m);
    printf("Cumulative TC : %d\n", cumulative_TC); printf("CGPA : %.2f\n", CGPA);
    printf("===== \n");
    return 0;
}

```

Test Cases

Test Case	Category	Inputs (global_id, m, persemester n; per-course code, credit, grade	Expected result	Actual result
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1	Valid / Normal	global_id=101, m=2 Sem1: n=2 → (101,4,O) (102,3,A) Sem2: n=1 → (201,4,B)	Sem1 SGPA=9.57 Sem2 SGPA=8.00 CGPA=9.00	Sem1 SGPA=9.57 Sem2 SGPA=8.00 CGPA=9.00 ✓
2	Boundary (minimum valid values	global_id=1, m=1 Sem1: n=1 → (1,1,P)	SGPA=4.00 CGPA=4.0	SGPA=4.00 CGPA=4.00 ✓
3	Invalid → corrected (negative / zero / bad grade)	global_id=-5→202 (re-enter) m=0→1 (re-enter) n=-2→1 (reenter) code=0→301, credit=3→3, grade=X→O (re-enter)	Each invalid entry triggers an error + reprompt; final SGPA=10.00, CGPA=10.00	All errors shown correctly; SGPA=10.00, CGPA=10.00 ✓

Program Execution (Linux Terminal Output)

```
kali@kali: ~/hdoc
(kali@kali) - [~/hdoc]
$ ./sgpa_cgpa
Enter Global ID: 101
Enter number of semesters (m): 2

----- Semester 1 -----
Enter number of courses (n) for semester 1: 2

Course 1 of Semester 1:
  Enter course code: 101
  Enter credit: 4
  Enter grade (0,A,B,C,D,E,P): 0

Course 2 of Semester 1:
  Enter course code: 102
  Enter credit: 3
  Enter grade (0,A,B,C,D,E,P): A

Semester 1 SGPA = 9.57

----- Semester 2 -----
Enter number of courses (n) for semester 2: 1

Course 1 of Semester 2:
  Enter course code: 201
  Enter credit: 4
  Enter grade (0,A,B,C,D,E,P): B

Semester 2 SGPA = 8.00

===== RESULT =====
Global ID      : 101
Semesters (m) : 2
Cumulative TC  : 11
CGPA           : 9.00
=====
```

```
kali@kali: ~/hdoc

(kali@kali)-[~/hdoc]
└─$ ./sgpa_cgpa
Enter Global ID: -5
Error: Global ID must be positive. Please re-enter.
Enter Global ID: 1
Enter number of semesters (m): 0
Error: Number of semesters must be positive. Please re-enter.
Enter number of semesters (m): 1

----- Semester 1 -----
Enter number of courses (n) for semester 1: -2
Error: Number of courses must be positive. Please re-enter.
Enter number of courses (n) for semester 1: 1

Course 1 of Semester 1:
Enter course code: 0
Error: Course code must be positive. Please re-enter.
Enter course code: 1
Enter credit: -3
Error: Credit must be positive. Please re-enter.
Enter credit: 1
Enter grade (0,A,B,C,D,E,P): X
Error: Invalid grade. Please re-enter.
Enter grade (0,A,B,C,D,E,P): P

Semester 1 SGPA = 4.00

===== RESULT =====
Global ID      : 1
Semesters (m) : 1
Cumulative TC  : 1
CGPA           : 4.00
=====
```